

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Complete

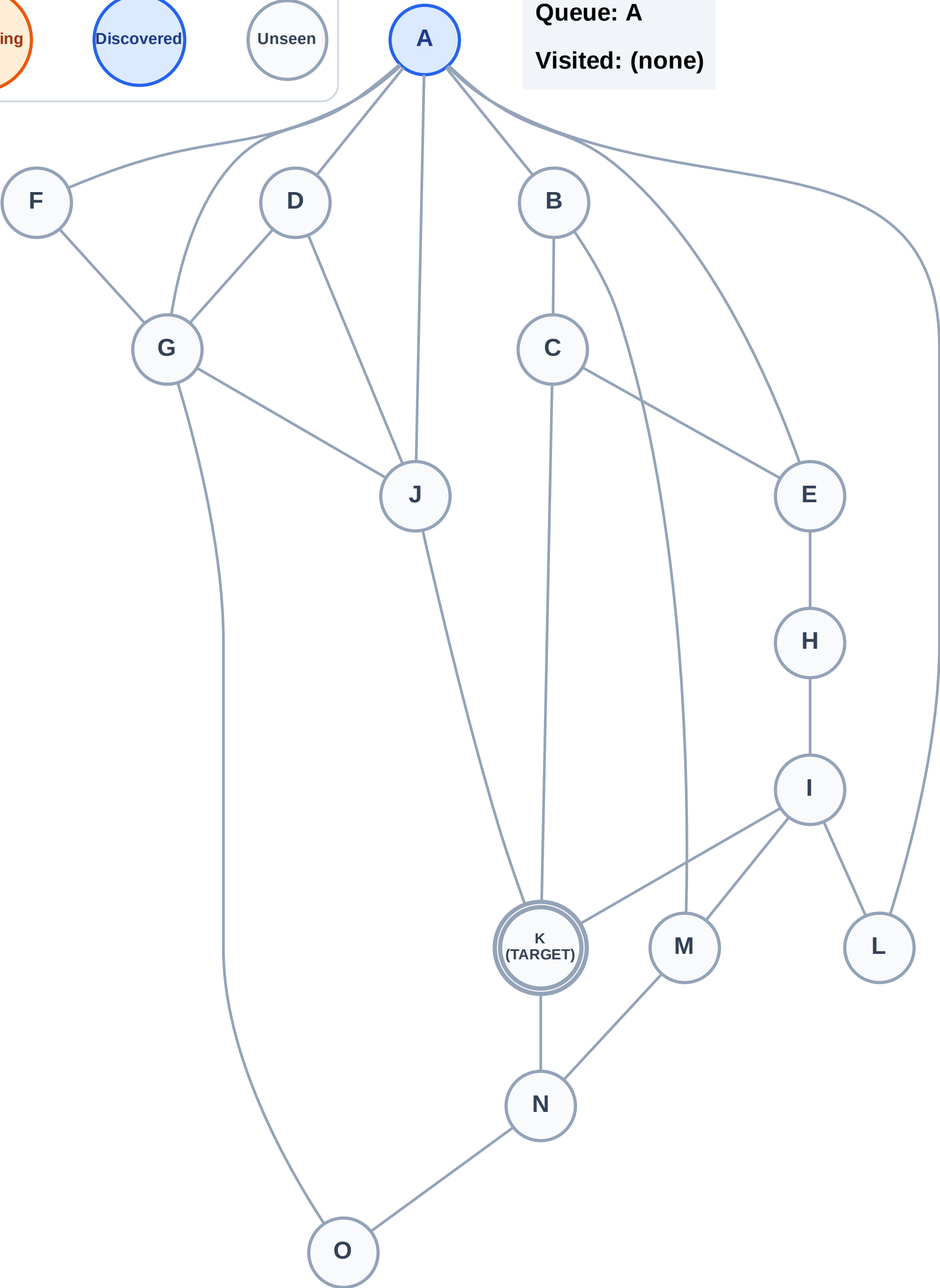
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

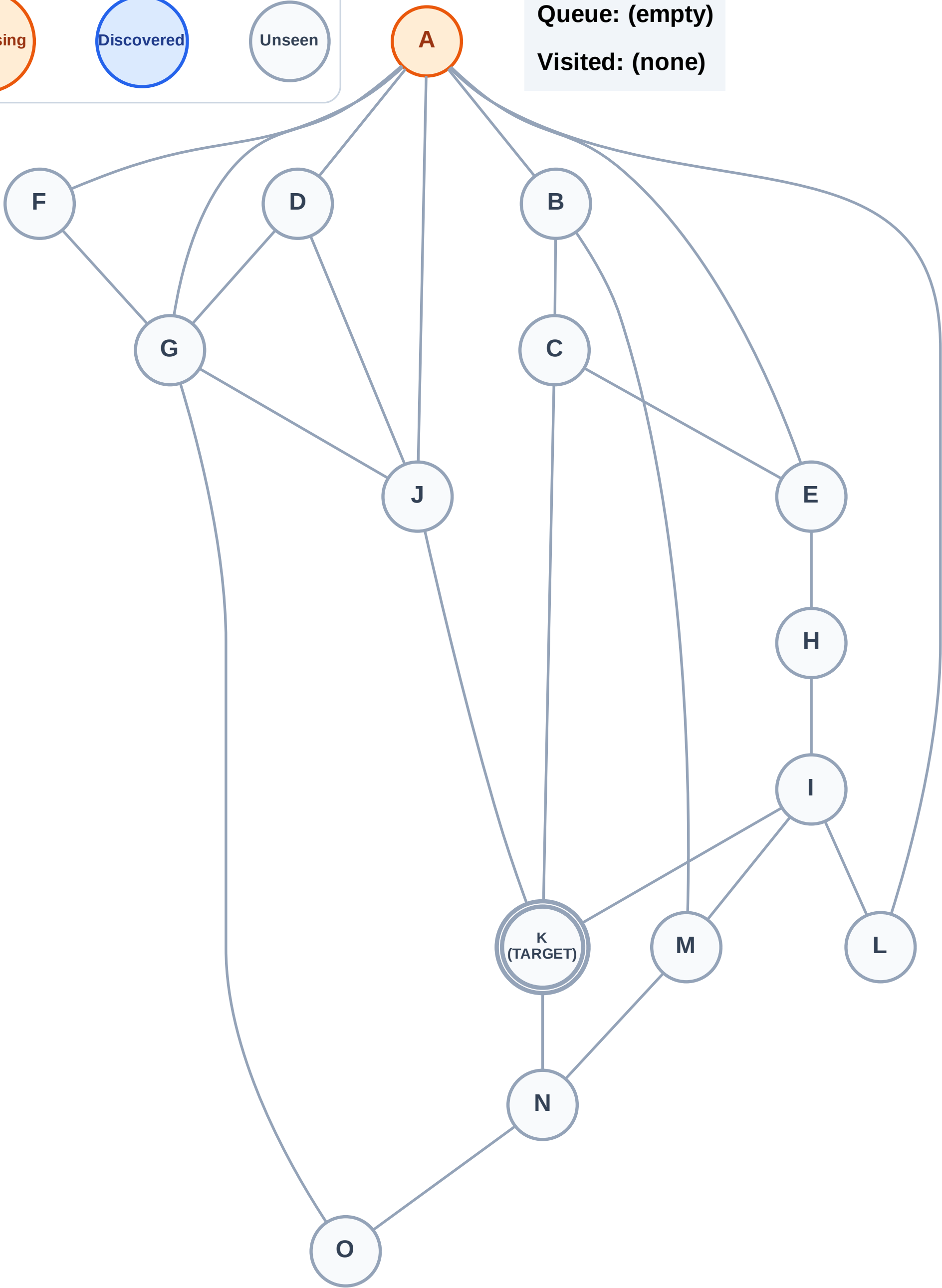
Complete

Processing

Discovered

Unseen

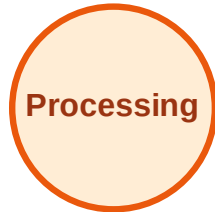
Queue: (empty)
Visited: (none)



BFS Traversal

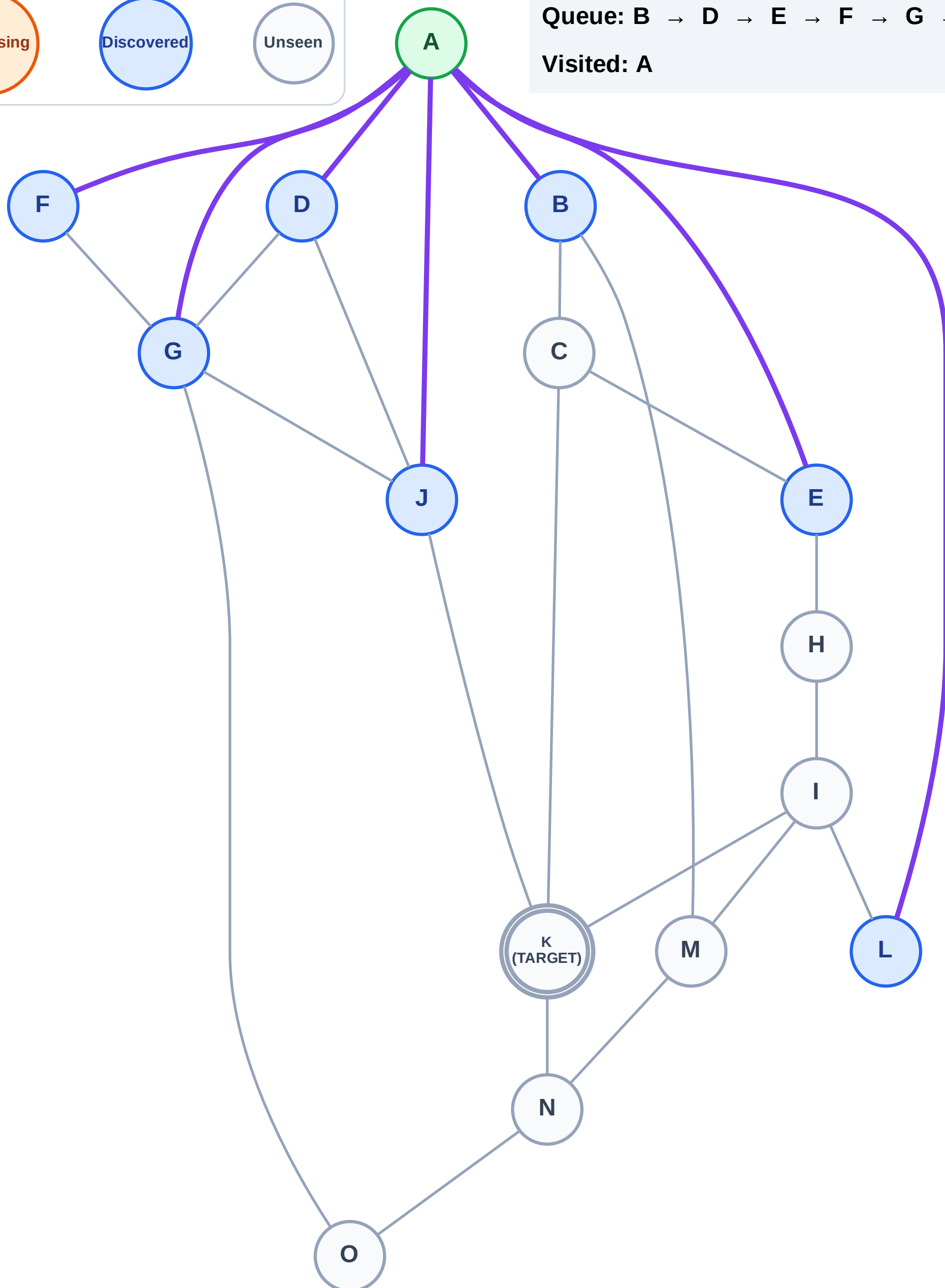
Step 3: Discover B, D, E, F, G, J, L from A

Legend



Queue: B → D → E → F → G → J → L

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

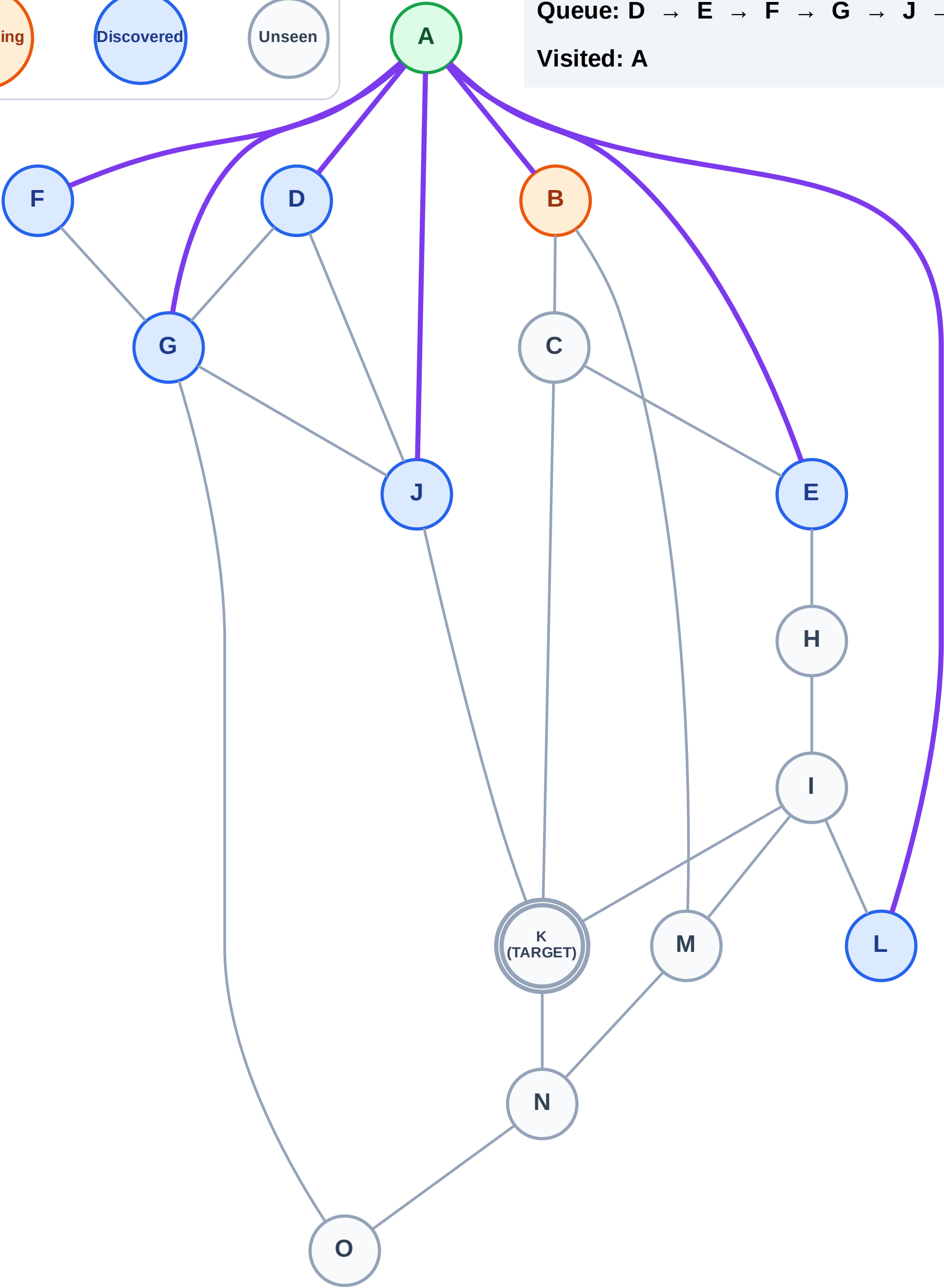
Processing

Discovered

Unseen

Queue: D → E → F → G → J → L

Visited: A



BFS Traversal

Step 5: Discover C, M from B

Legend

Complete

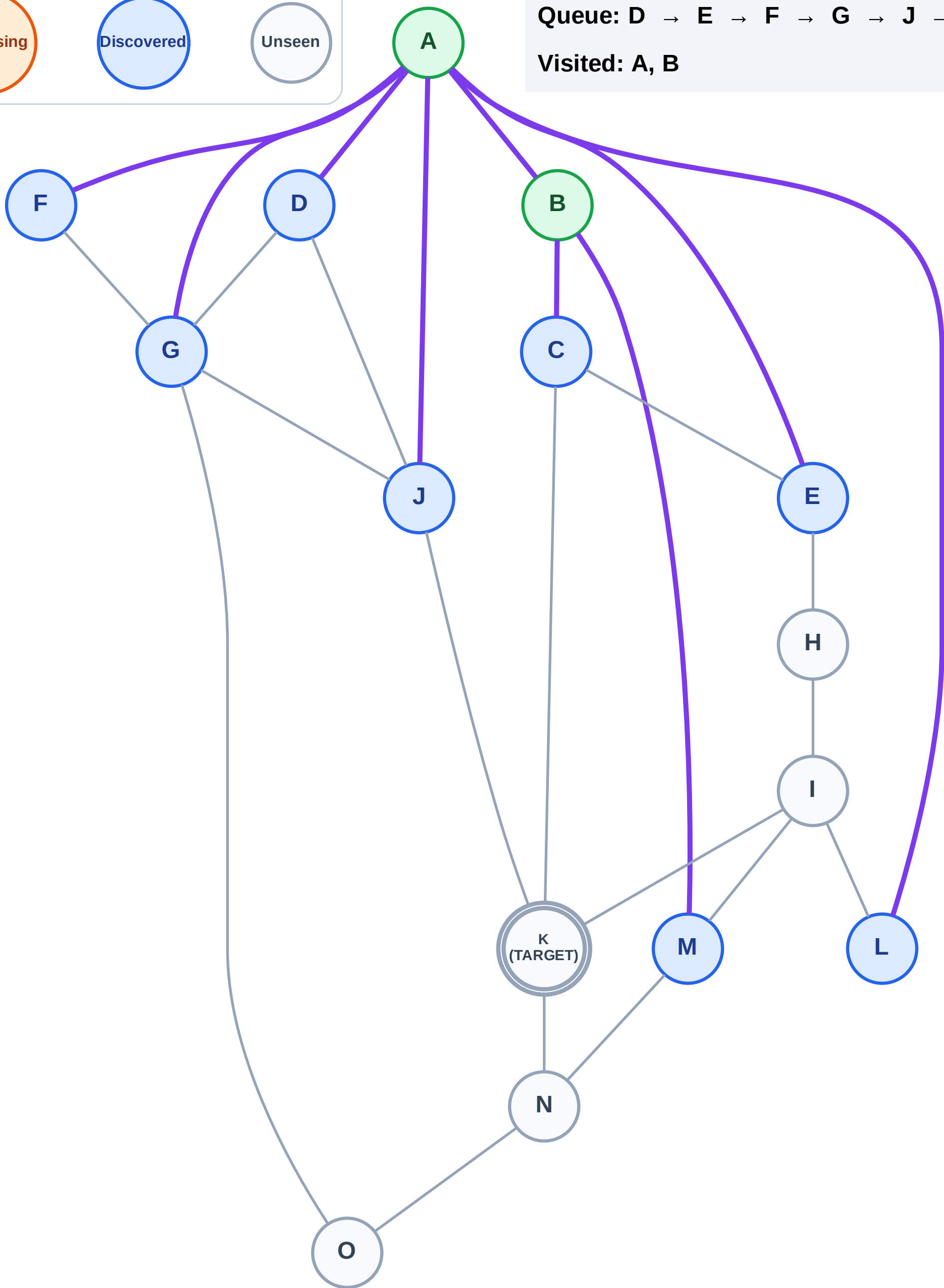
Processing

Discovered

Unseen

Queue: D → E → F → G → J → L → C → M

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

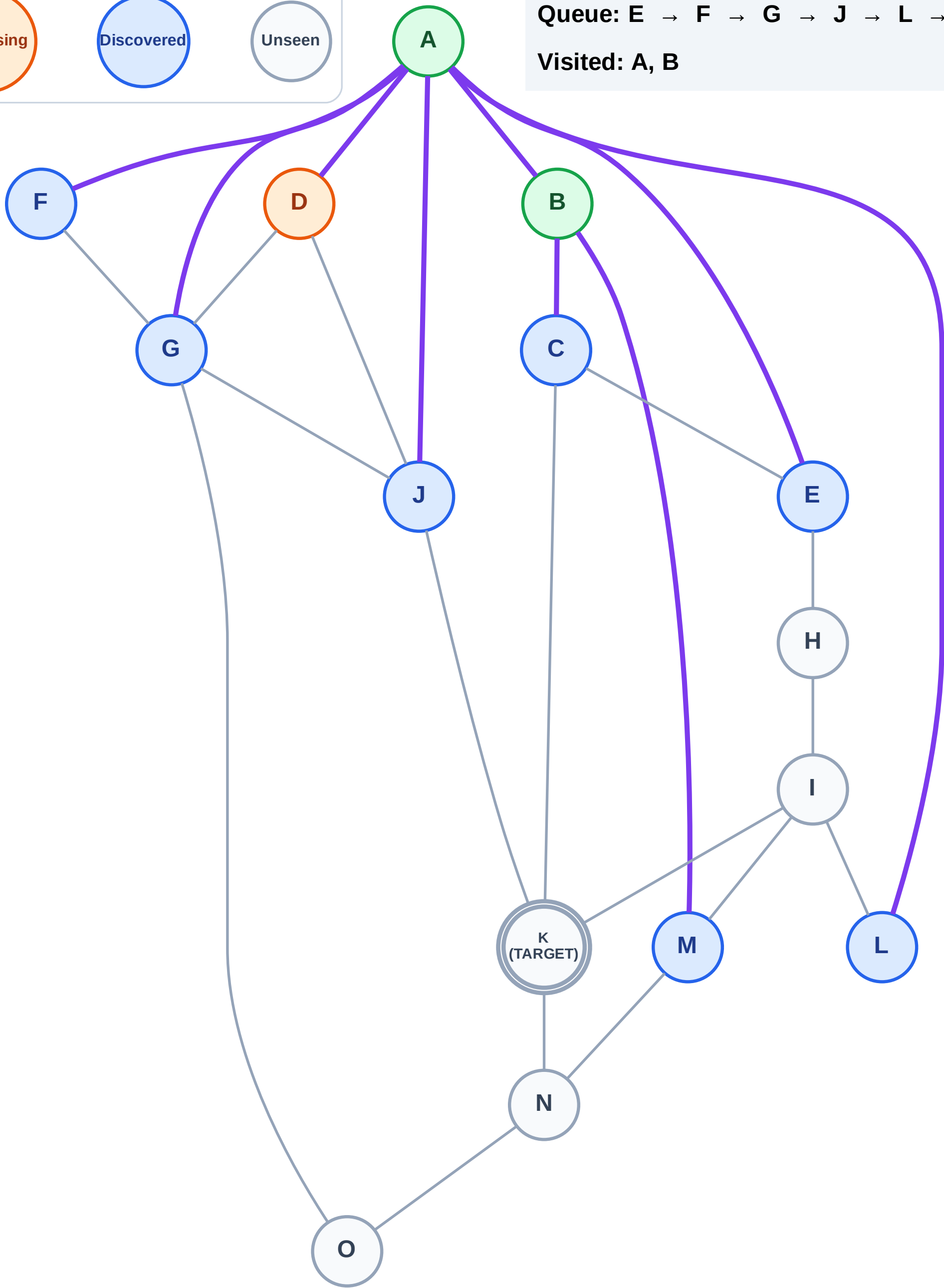
Processing

Discovered

Unseen

Queue: E → F → G → J → L → C → M

Visited: A, B



BFS Traversal

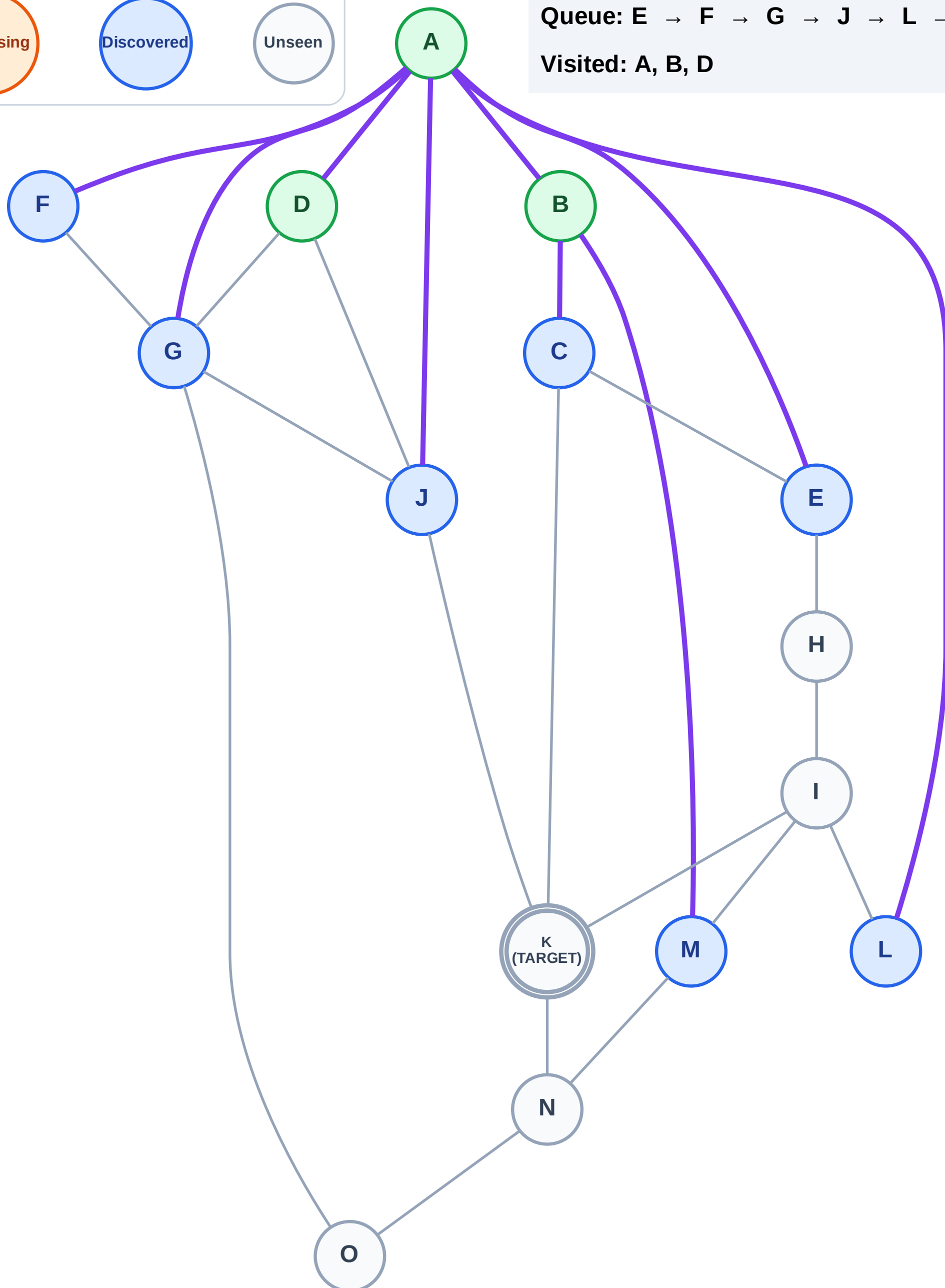
Step 7: No new neighbors from D

Legend



Queue: E → F → G → J → L → C → M

Visited: A, B, D



BFS Traversal

Step 8: Process node E

Legend

Complete

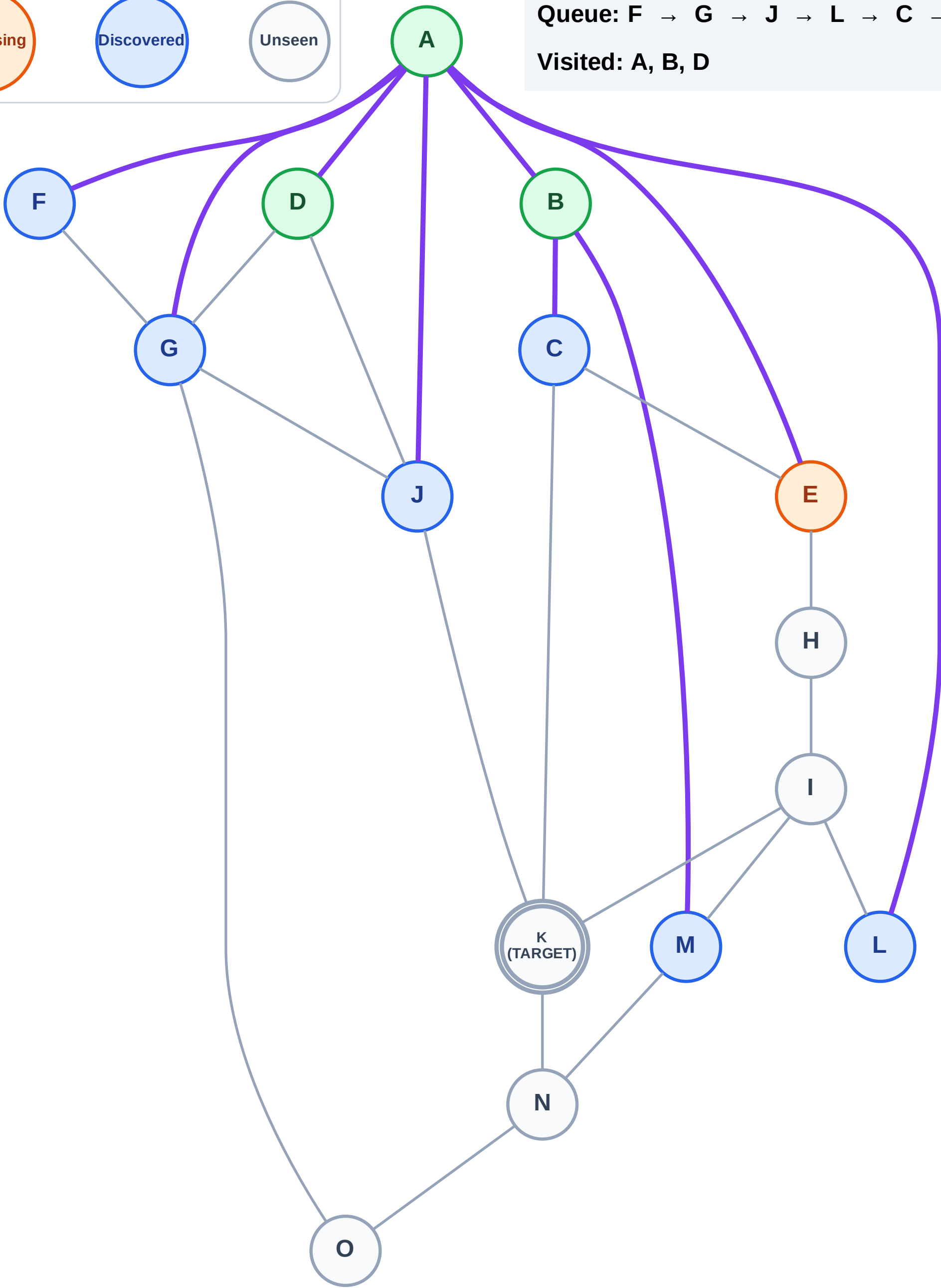
Processing

Discovered

Unseen

Queue: F → G → J → L → C → M

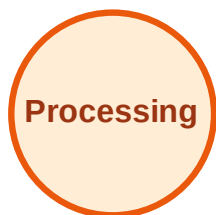
Visited: A, B, D



BFS Traversal

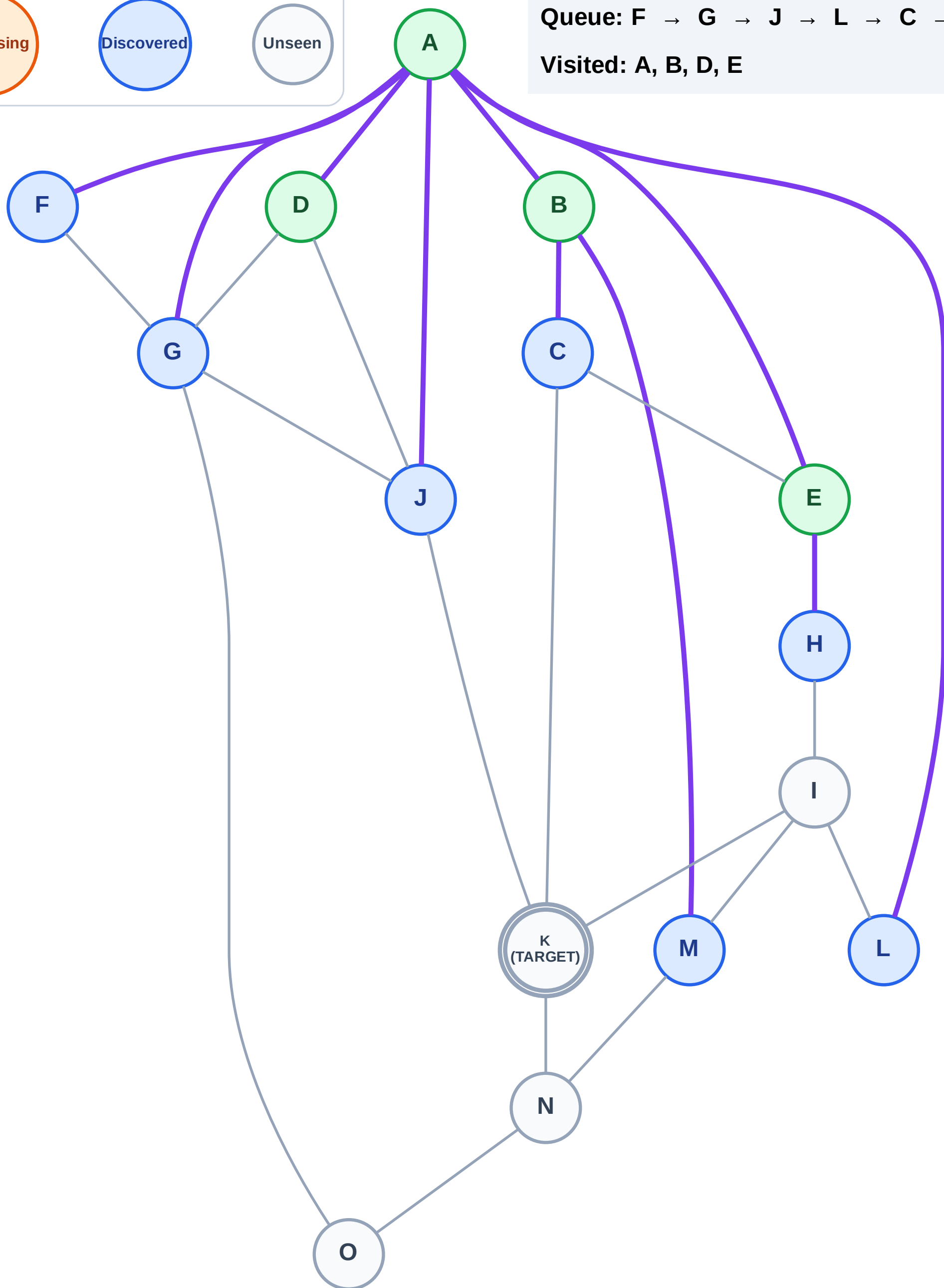
Step 9: Discover H from E

Legend



Queue: F → G → J → L → C → M → H

Visited: A, B, D, E



BFS Traversal

Step 10: Process node F

Legend

Complete

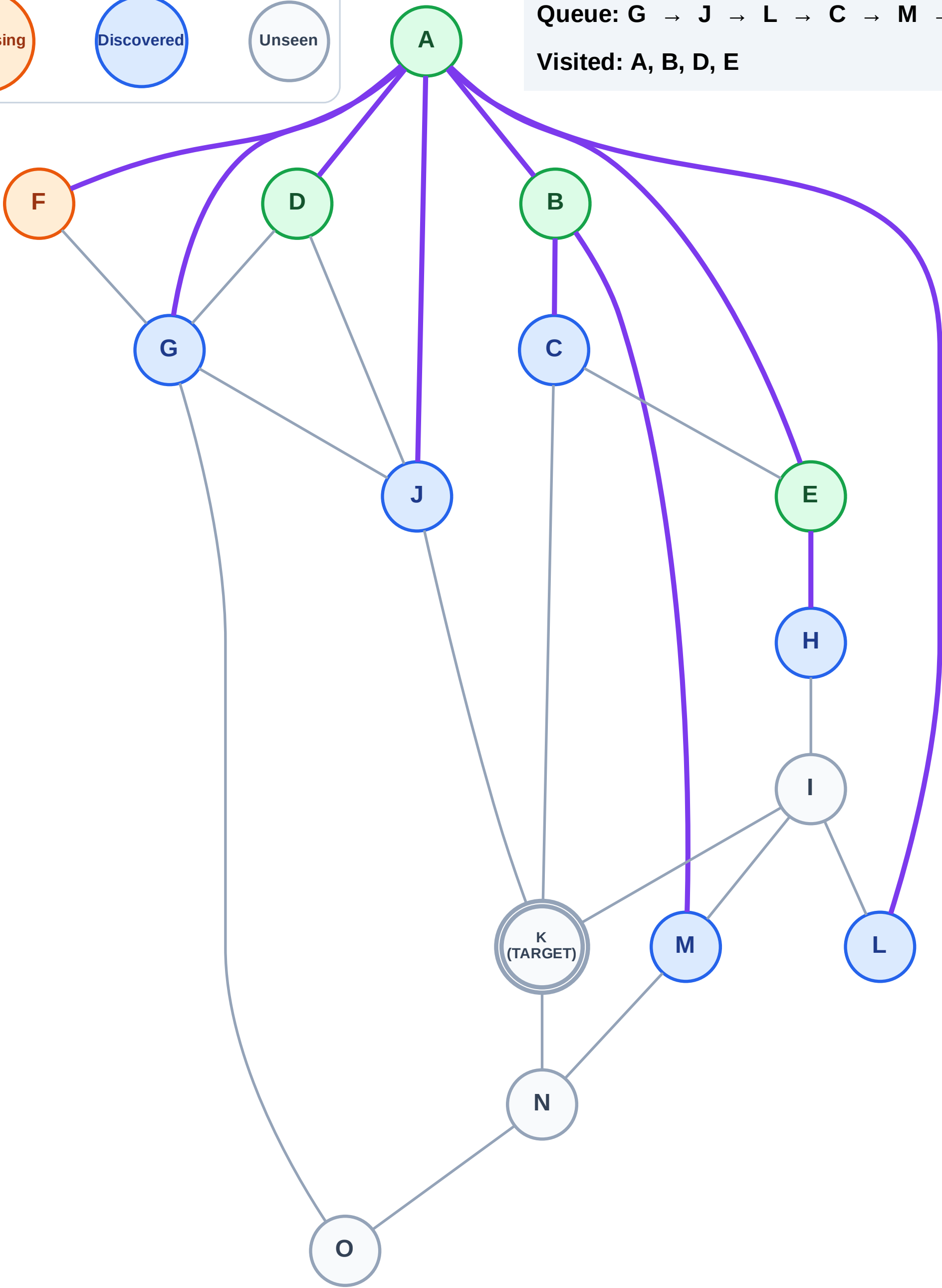
Processing

Discovered

Unseen

Queue: G → J → L → C → M → H

Visited: A, B, D, E



BFS Traversal

Step 12: Process node G

Legend

Complete

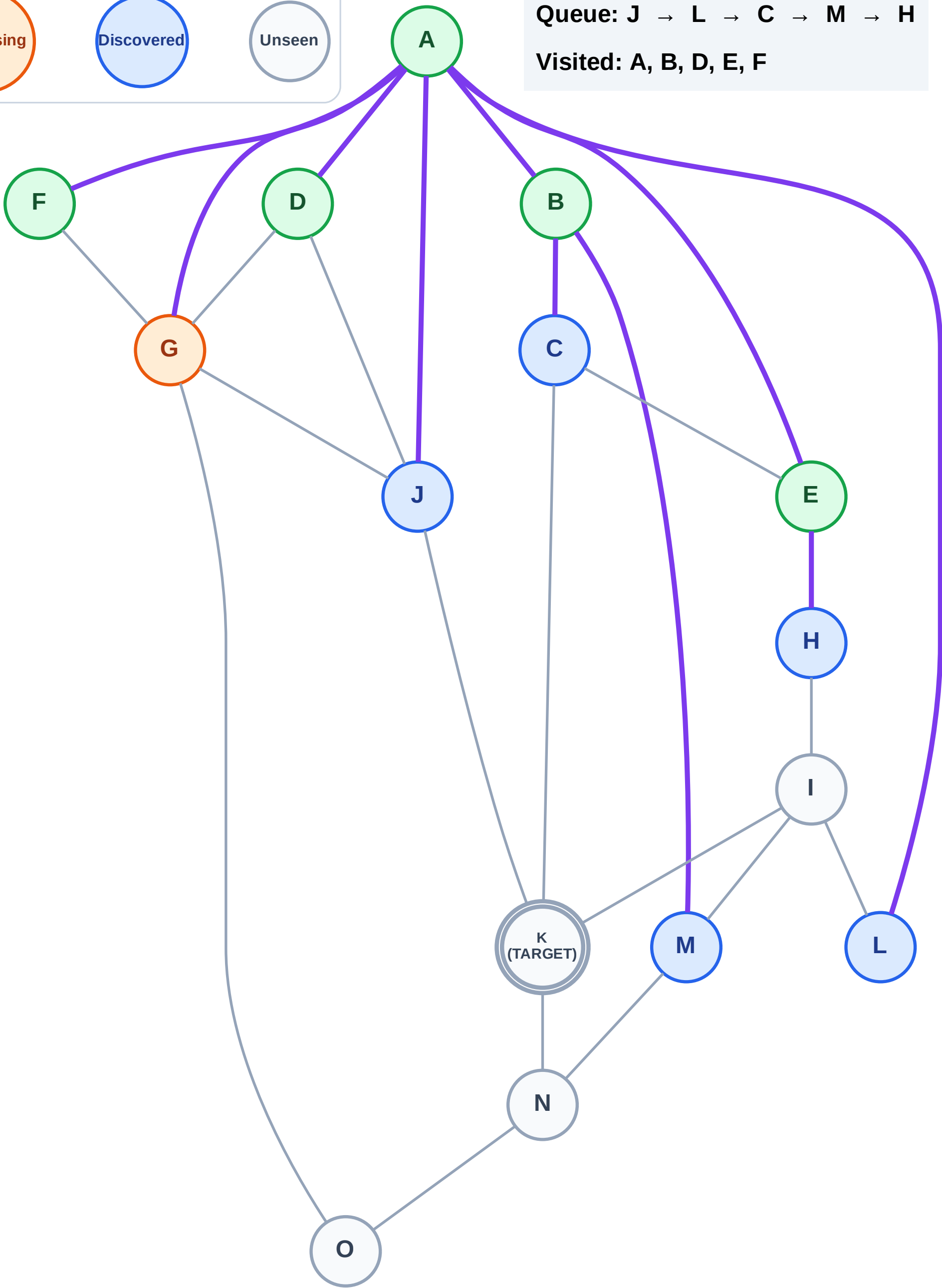
Processing

Discovered

Unseen

Queue: J → L → C → M → H

Visited: A, B, D, E, F



BFS Traversal

Step 13: Discover O from G

Legend

Complete

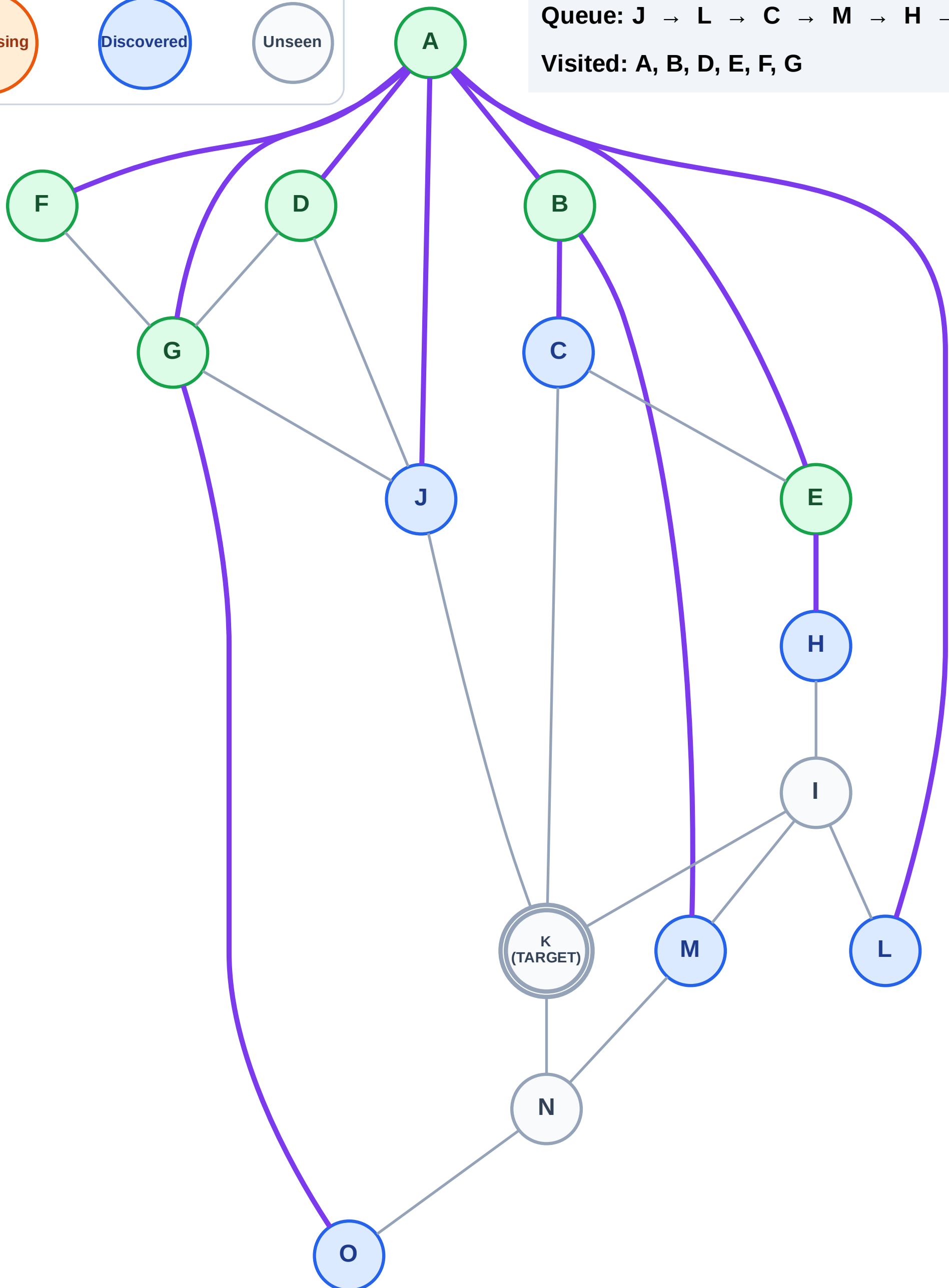
Processing

Discovered

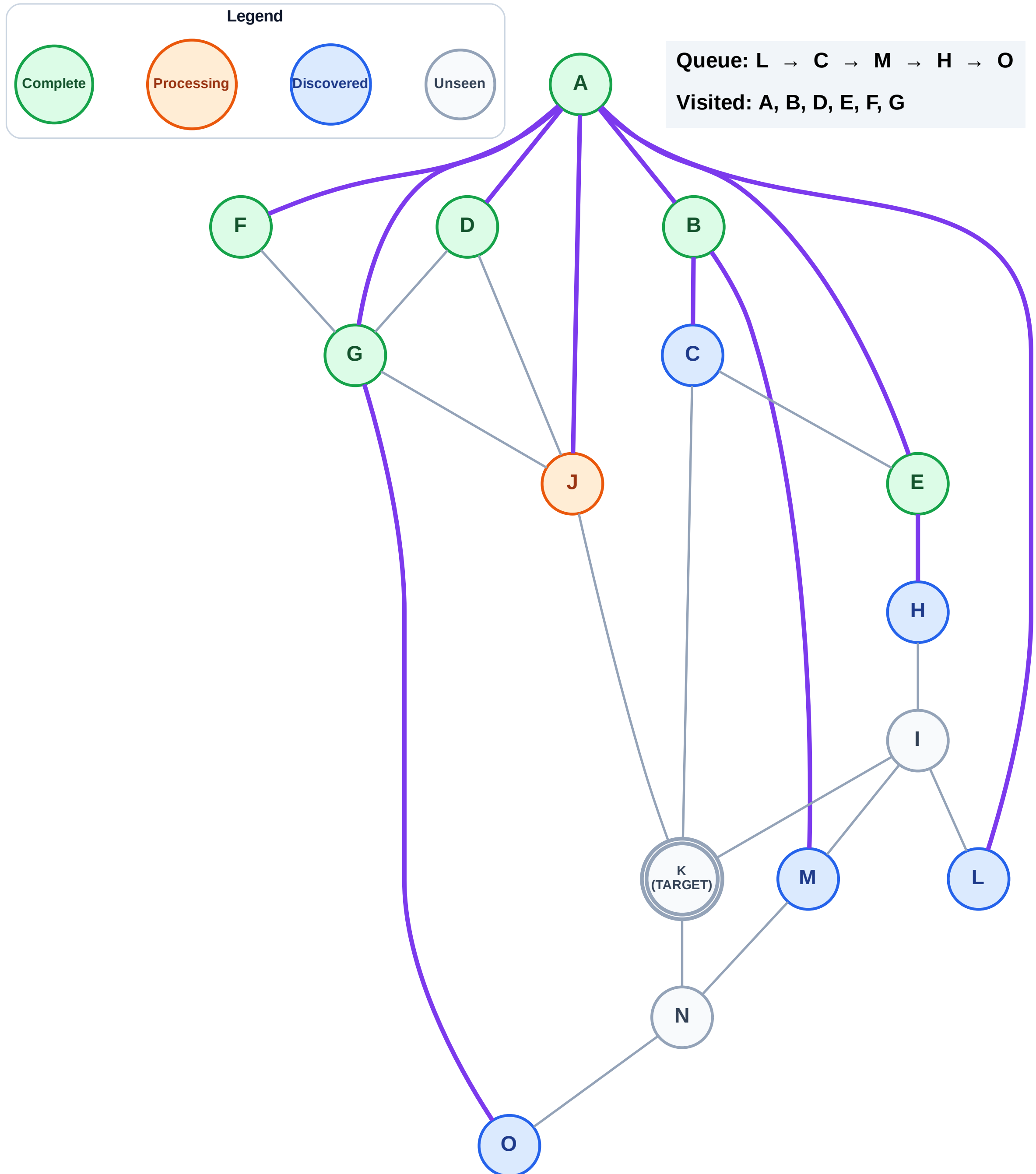
Unseen

Queue: J → L → C → M → H → O

Visited: A, B, D, E, F, G



Step 14: Process node J



BFS Traversal

Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → J → K

Legend

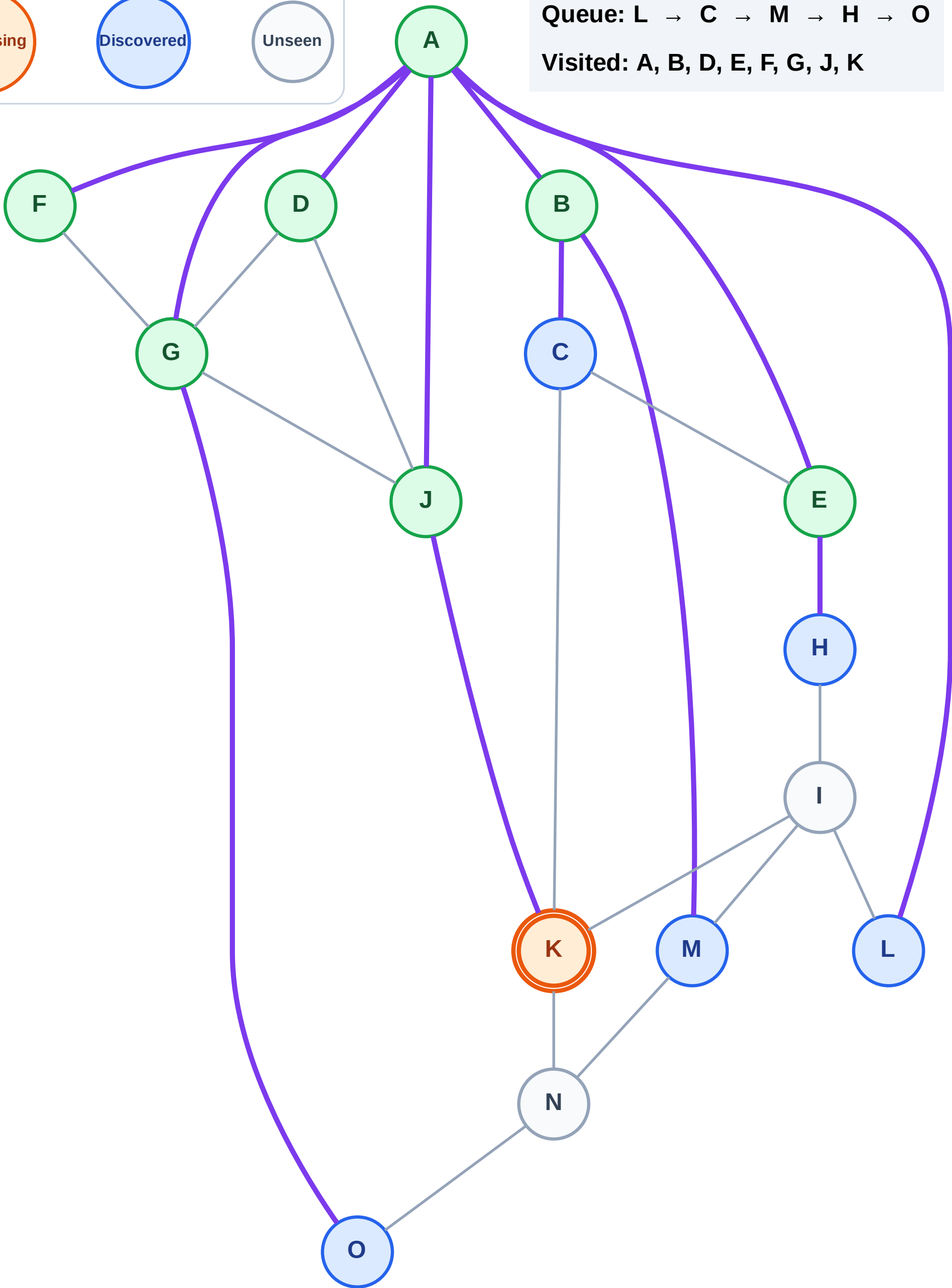
Complete

Processing

Discovered

Unseen

Queue: L → C → M → H → O
Visited: A, B, D, E, F, G, J, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

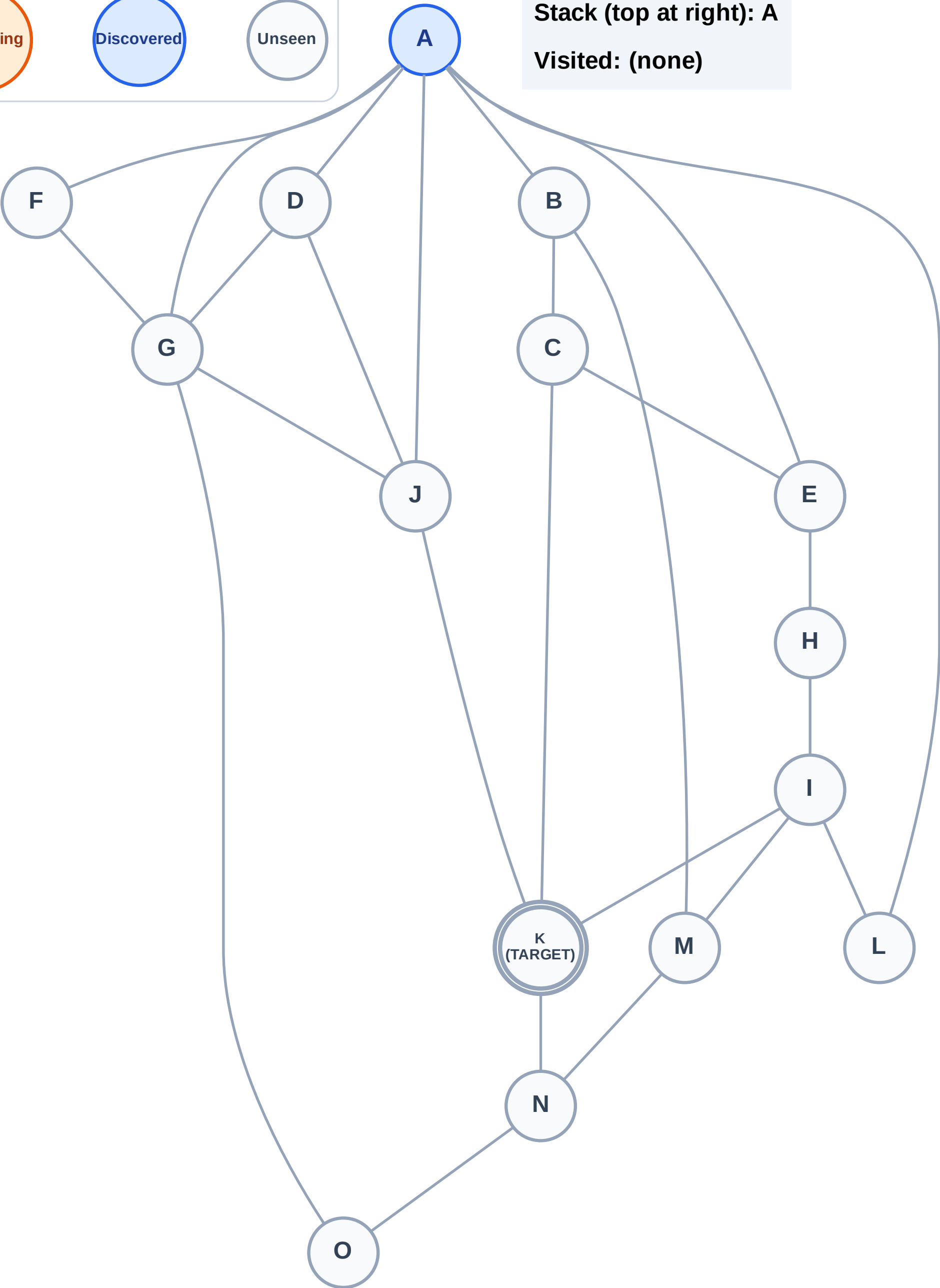
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

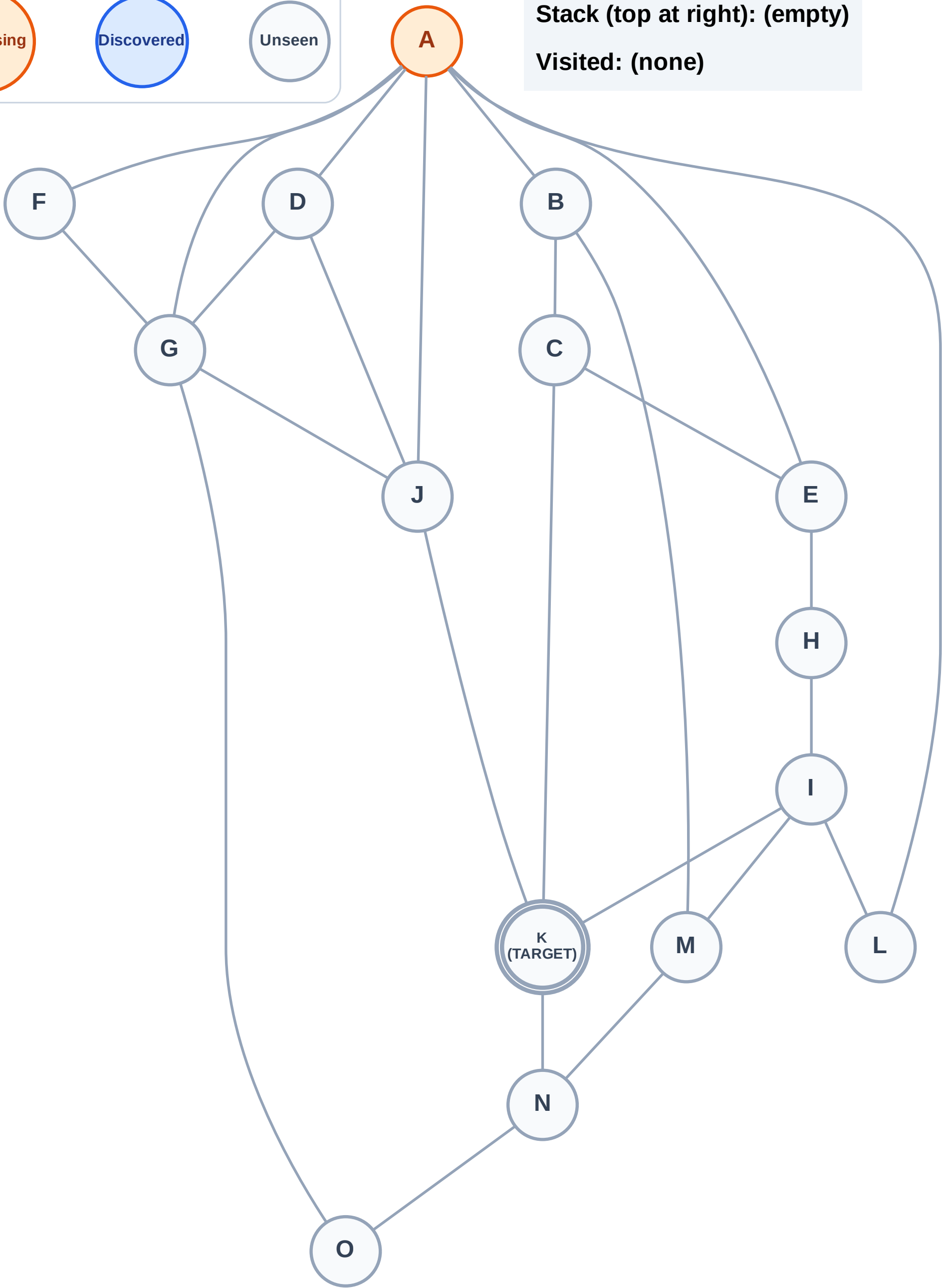
Complete

Processing

Discovered

Unseen

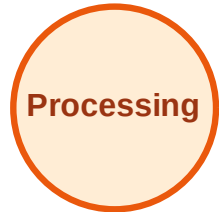
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

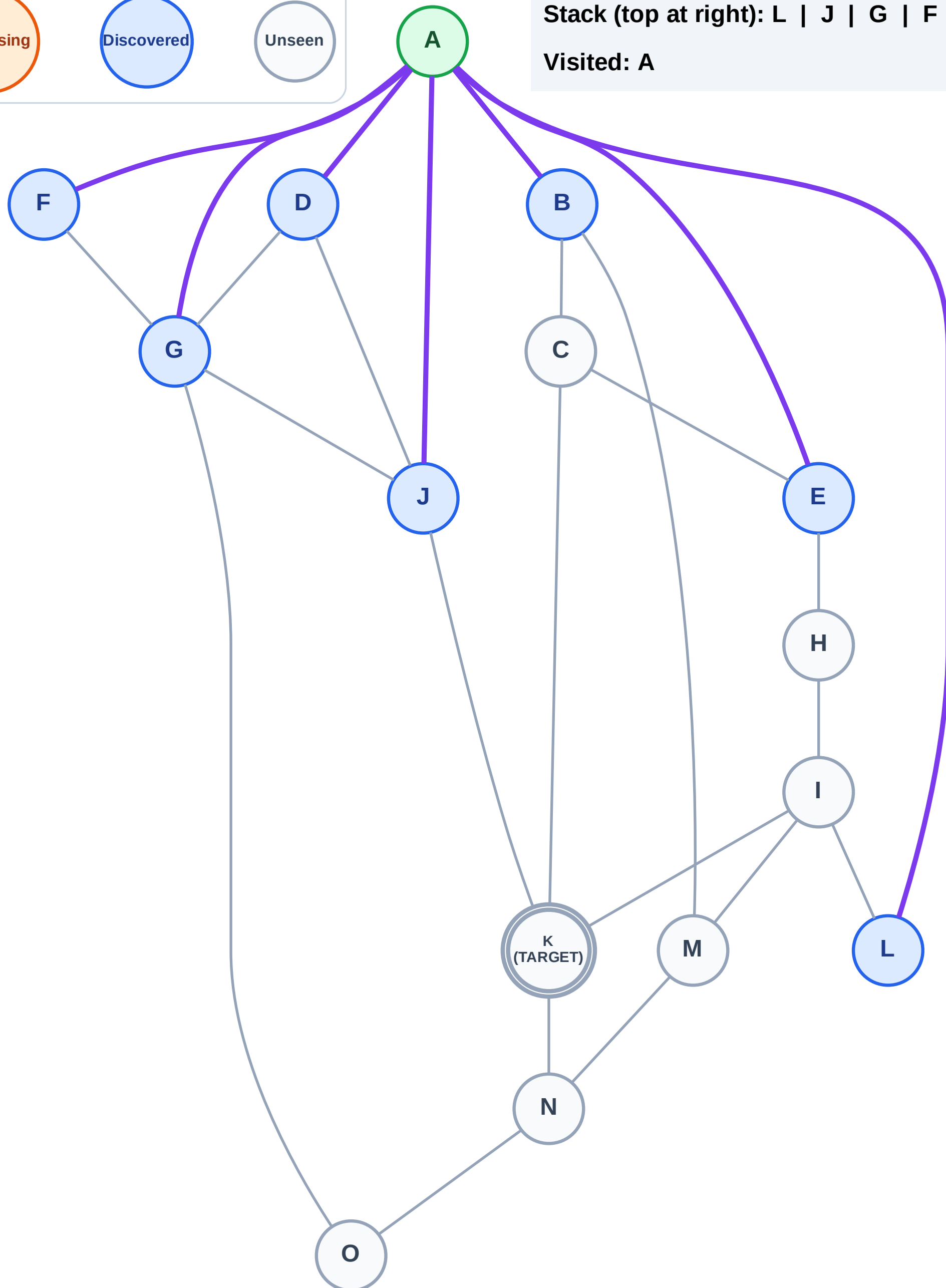
Step 3: Discover L, J, G, F, E, D, B from A

Legend



Stack (top at right): L | J | G | F | E | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

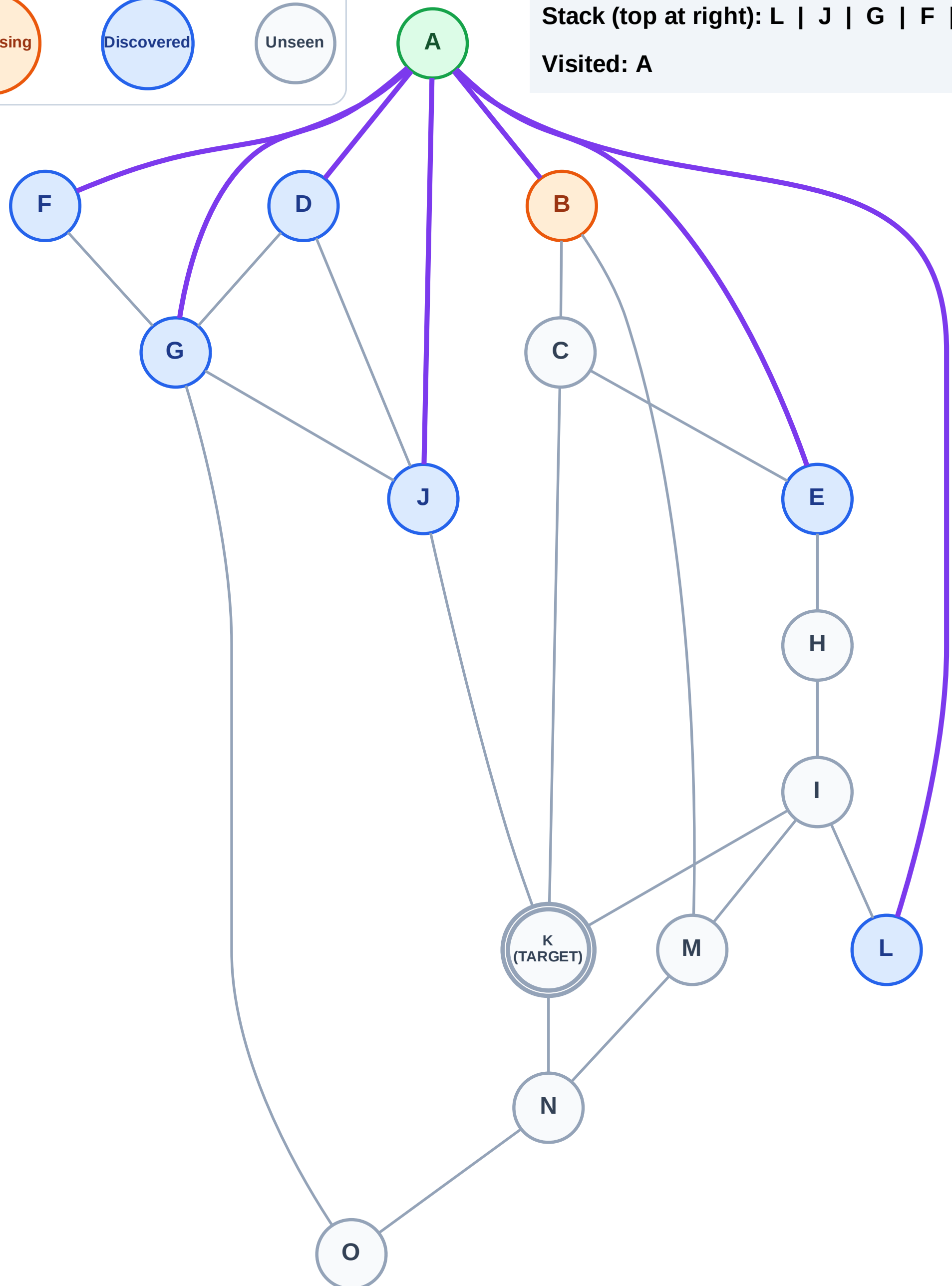
Processing

Discovered

Unseen

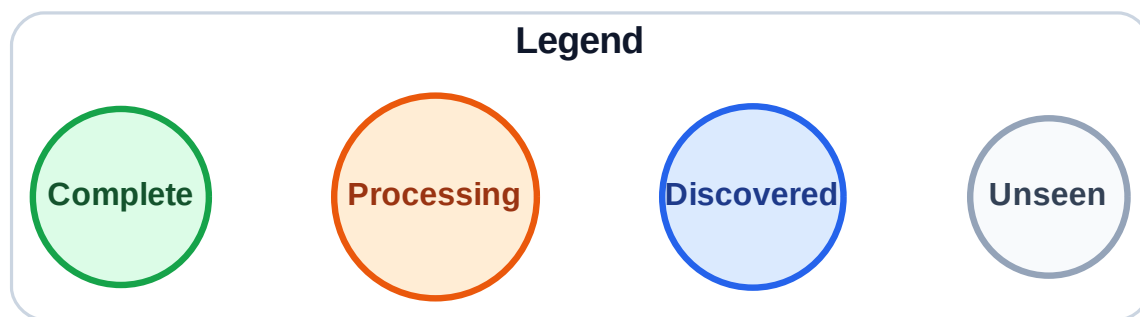
Stack (top at right): L | J | G | F | E | D

Visited: A



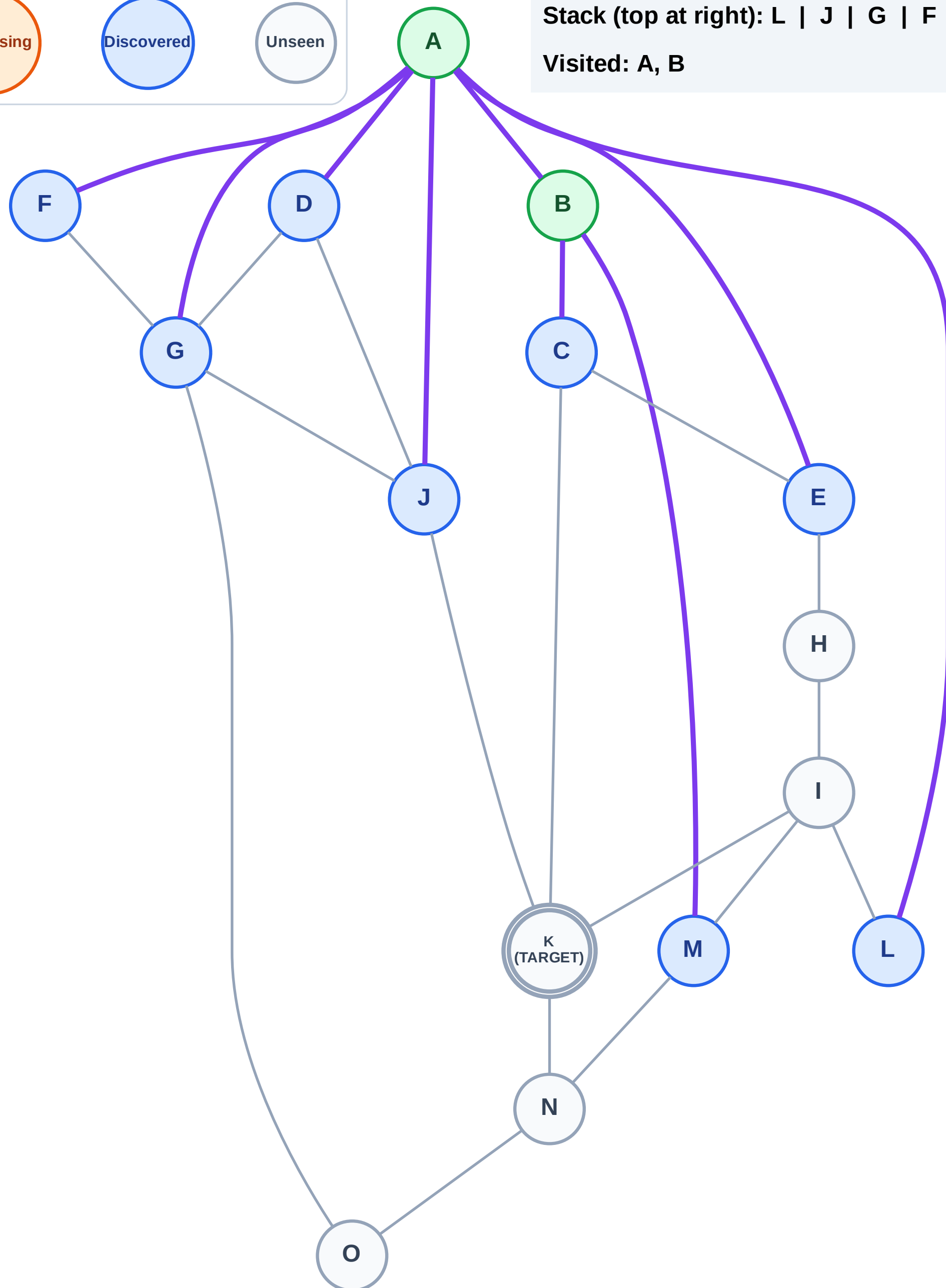
DFS Traversal

Step 5: Discover M, C from B



Stack (top at right): L | J | G | F | E | D | M | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

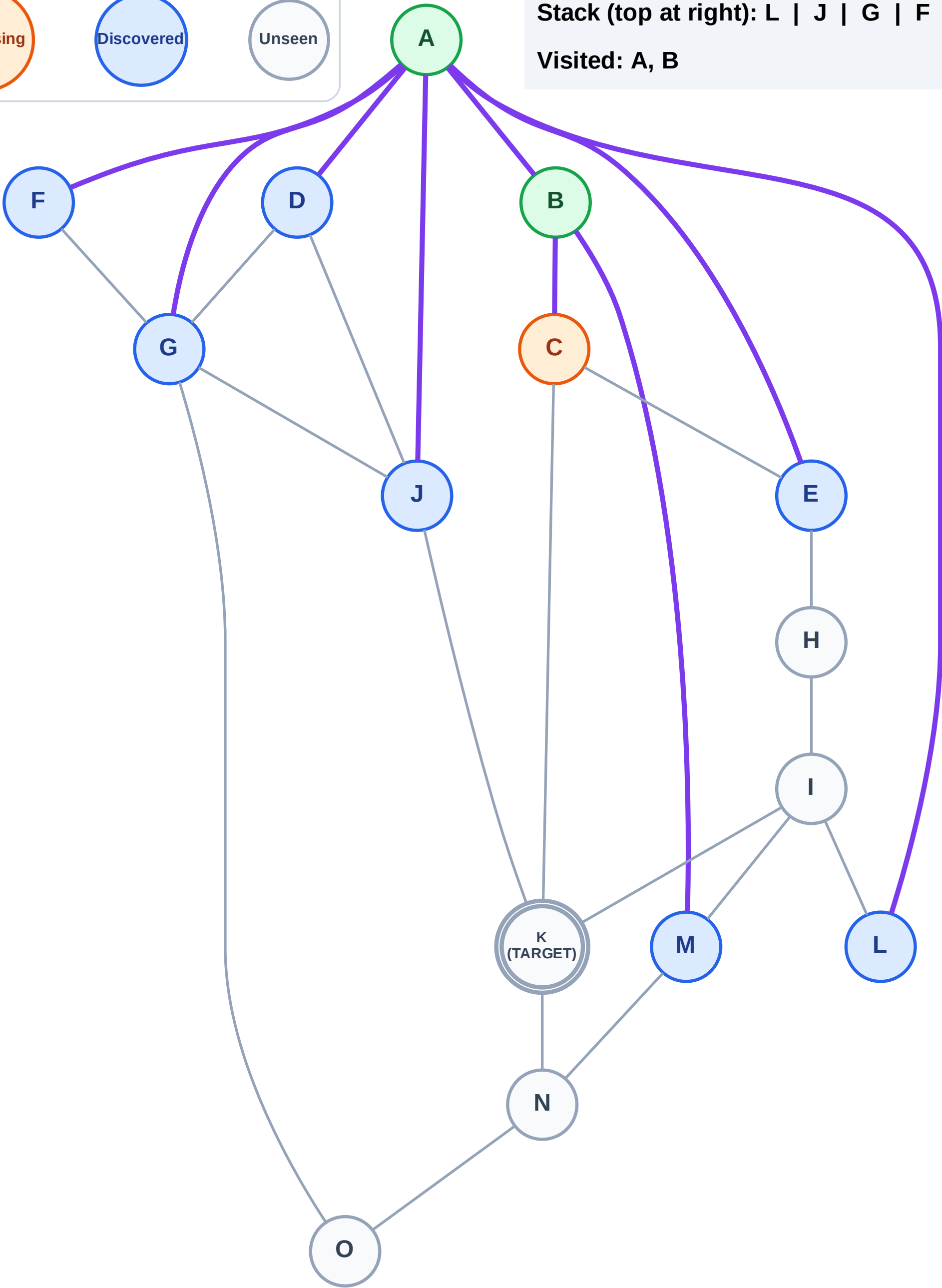
Processing

Discovered

Unseen

Stack (top at right): L | J | G | F | E | D | M

Visited: A, B



DFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → C → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): L | J | G | F | E | D | M

Visited: A, B, C, K

